

IEIH ESPORTS APPLICATION

Architecture, Tech Stack, & Documentation Overview

1. Purpose of the Application

The IEIH Esports Hub is designed to streamline esports team management and talent acquisition. It provides a centralized platform connecting players, talent scouts, and coaches through modern web and mobile channels. Key functionalities include:

- Player Scouting & Recruitment:

Allows talent scouts and managers to view active player rosters and submit direct scouting offers to promising candidates.

- EEAI Tactical Chatbot:

An integrated AI assistant providing real-time gameplay tactical analysis and strategy suggestions for players and coaches.

- Multi-Platform Support:

Accessible via a responsive web interface (Netlify) and a native mobile application (Android/iOS via Expo).

2. Project Structure

```
ieih-app/
|-- backend/                # Node.js + Express + TypeScript Backend API
|   |-- src/                # Source code (server, routes, controllers)
|   |-- tsconfig.json       # TypeScript configuration
|   |-- package.json        # Backend dependencies & scripts
|
|-- frontend/               # React + Vite + TypeScript Frontend Client
|   |-- src/                # Client source code (components, API services)
|   |-- src/services/api.ts  # API communication service (Netlify/Local switch)
|   |-- tailwind.config.js   # Tailwind CSS styling configuration
|   |-- package.json        # Frontend dependencies & scripts
|
|-- mobile/                 # Expo (React Native) Mobile Application
|   |-- App.tsx              # Mobile entry point and UI screens
|   |-- app.json             # Expo project configuration
|   |-- package.json        # Mobile dependencies & scripts
|
|-- package.json            # Root-level configuration
|-- README.md               # Documentation
```

3. Technology Stack & Purpose

Component	Technology	Purpose
Monorepo	Node.js	Unified project management and scripts.
Backend	Express	Handles REST API endpoints (health, players, scout, AI chat).
Backend	TypeScript / tsx	Type-safety and modern ES module execution.
Backend	CORS & dotenv	Enables Cross-Origin requests and secures configuration.
Frontend	React 19	UI component architecture and state management.
Frontend	Vite	Ultra-fast local development server and bundler.
Frontend	Tailwind CSS	Utility-first styling for responsive layouts.
Frontend	Lucide React	Modern, lightweight iconography.
Mobile	Expo / React Native	Single codebase for compiling iOS & Android mobile apps.
Mobile	EAS Build	Cloud compilation of release binaries (.apk, .aab, .ipa).
Deployment	Netlify	Automated continuous integration and web hosting.
Deployment	Render	Managed cloud hosting for the continuous Express backend service.

4. Quick Start Summary

To run components locally, execute the following commands in respective folders:

Backend Server: `cd backend && npm install && npm run dev`
Frontend Client: `cd frontend && npm install && npm run dev`
Mobile App: `cd mobile && npm install && npx expo start`